

Yiming Cao

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EDUCATION

University of British Columbia — M.Eng., Electrical & Computer Engineering Sept 2022 – May 2024 · Vancouver, BC

- GPA 87%. Focus: deep learning, computer vision, generative models, medical image analysis, reinforcement learning.

University of Victoria — B.Sc. with Distinction, Honours Combined Computer Science and Mathematics Sept 2017 – Apr 2021 · Victoria, BC

- Cumulative GPA 7.53/9. Honours thesis on off-policy reinforcement learning (supervisor: Nishant Mehta). Faculty of Science Dean's List (2020); International Student Scholarship (2017).

EXPERIENCE

Machine Learning Engineer — Xi'an University of Posts and Telecommunications (remote) Sept 2026 – present
Embodied AI for robotic parcel sorting with [TWOVIN Technology](#), a Shenzhen AI hardware company.

- Robotic arm with grippers for parcel sorting, supervised by Dr. Chen Wei: a pre-trained vision-language-action / world-action model, post-trained on human teleoperation data collected on the robot. Currently onboarding the post-training pipeline ahead of data collection.

Software Engineer (volunteer) — Tomkulak Consortium (remote) Jan 2026 – present
Critical-infrastructure system-modelling platform: 21 sectors, Vancouver Island MVP scaling to BC, Yukon and Alberta; three layers — data ingestion, system modelling, visualization. Async Agile team with PR-based code review.

- Engineered the configuration-driven **Python** ingestion template adopted by all 21 sector pipelines — pluggable **Shapefile / GeoPackage / File Geodatabase / ArcGIS** readers, staging → validation → transformation (**GeoPandas, Shapely**) → **PostgreSQL/PostGIS** load, per-sector **YAML** configs, 186 tests — and its validation framework (**Pydantic, SQLAlchemy**), whose reports separate code bugs from data problems to send back to the GIS team; 192 tests.
- Owned the system-modelling architecture: the nine-entity object model, a **NetworkX** dependency graph (cascading impact, bottlenecks, build order, redundancy groups), and the versioned **Pydantic** Stage 1 → Stage 2 data contract that let modelling and the **React** frontend proceed in parallel with ingestion.
- Built the dependency and constraint framework (eight relationship categories as typed graph edges; feasibility rules in editable **YAML**) and the scenario framework (side-by-side comparison of 3+ configurations, sensitivity analysis over six variables, typed **JSON** results matched to the frontend's endpoints).
- Built the multi-objective scoring and ranking engine: eight bounded objectives, **YAML** weight profiles, Pareto flags, deterministic rankings, and explanation metadata on every score reconstructing it to 1e-9; 253 modelling tests, byte-identical output verified on the pilot.

Software Engineer — NextTier, Covina, CA Sept 2024 – Sept 2025
Designed and built from scratch a music-focused RAG chatbot for a client — backend, retrieval layer, UI and AWS deployment.

- Built the **Java 17 / Spring Boot** backend (REST APIs for message flow, chat history, session state and tone adjustment over **MongoDB**) and the **RAG** retrieval layer — **OpenAI text-embedding-3** with **k-NN vector search** on **MongoDB Atlas** — grounding **GPT-4o** responses in the client's verified lyric and music corpus, enriched from **HuggingFace** datasets and a chunked analysis of a ~9 GB lyrics dataset.
- Delivered a **React + TypeScript + Tailwind CSS** chat interface with end-to-end token streaming and context retention, iterated over 4 months of client feedback; built **Kafka** producer/consumer event services alongside the chat backend.
- 100% test coverage with **JUnit** and **Mockito**; API results published in **Postman**; deployed to **AWS Elastic Beanstalk** with autoscaling and **CloudWatch** monitoring.

Research Intern — Xi'an University of Posts and Telecommunications, Xi'an, China May – Aug 2023
Researched denoising diffusion models for generating directed acyclic graphs as deep-network architectures (neural architecture search) with Dr. Chen Wei; literature review, code implementation, reports and presentations.

Research Intern — Path Academics (remote) Aug – Sept 2020
*Reinforcement-learning-based tracking for occlusion handling, supervised by Pietro Liò — modelled the tracking problem mathematically, implemented the deep RL model in **Python**, analysed it against real-world scenarios, presented and wrote the report.*

- Collaborative-filtering algorithms for recommendation systems, supervised by Yiqiang Sheng — modelled the approaches mathematically, implemented them in **Python**, evaluated them on real-world data, presented and wrote the report.

INDIVIDUAL PROJECTS

Job Hunting Assistant — multi-site job scraper with an LLM matching pipeline

Mar 2026 – present · [GitHub](#)

- **Chrome extension** (unattended **Manifest V3** service-worker orchestrator) scraping LinkedIn, Indeed and Glassdoor into a **FastAPI/PostgreSQL** backend with a **React** console — 1,306 postings ingested in one unattended cycle; 10-hour soak tests exposed and fixed race conditions.
- Dual-write data model (per-site append-only tables plus a 27-column canonical table, 31 **Alembic** migrations) and a filter-and-match service that puts deterministic gates ahead of **gpt-4o-mini**, so every run reconciles `claimed = shortlisted + excluded + anomalies` — ~70% fewer input tokens; 184 service tests, 135 frontend tests; built spec-first across 11 feature specs.

My Daily News — an AI front page that can prove it showed me everything

Sept 2026 – present · [GitHub](#)

- Reads 48 AI-focused sources each morning (33 feeds, 9 page scrapers, 6 ranking snapshots) through one adapter template, shows only unseen items, and summarises and categorises them with one **LLM** call per item (**Python**, **FastAPI**, **React** + **Vite**, **gpt-5.4-mini**); a 7-day window decides what is examined, a seen store keyed on (source, time, normalised title) decides what is shown.
- Completeness is checked, not assumed: an independent reconciliation re-parses raw feed bytes — 0 of 2,401 entries unaccounted; 261 tests. Hypothesis thresholds fixed before the data: LLM categorisation reached 83.1% against an 85% bar and stands as a fail; code frozen since Sept 28, 2026 for a five-morning measurement.

E-Commerce Admin Platform — Go REST API and React dashboard

Nov – Dec 2025 · [GitHub](#)

- 28-route **Go** (**Fiber v3** + **GORM/MySQL**) API and 14-route **React/TypeScript** dashboard; **JWT** cookie auth, **bcrypt**, role-based access control; written security review of the remaining gaps.

RESEARCH PROJECTS — UBC M.ENG. COURSEWORK

- **Detecting AI-generated images by their missing Bayer pattern** (EECE 541, Panos Nasiopoulos; Apr 2024) — a training-free detector using PSNR disagreement across the four Bayer layouts: 97–100% detection on 46,000 generated images vs 999 camera RAWs; batched **PyTorch** demosaicing and metrics; released the "Landscape PhotoReal" dataset.
- **Age-invariant face recognition by knowledge distillation** (EECE 571L, Panos Nasiopoulos; Apr 2024) — responsible for code and evaluation; distilled a ResNet-100 ElasticFace teacher into ElasticFace and MTLFace students on B3FD (**SLURM**, Compute Canada V100s); +19.5 points on AgeDB for a weak student.
- **Denoising Diffusion Probabilistic Models** (EECE 501, Lele Wang; Dec 2023) — derived the DDPM objective end to end; noise-schedule experiments in **PyTorch**.
- **Reinforcement learning for a Robocode AI tank** (CPEN 502, Sarbjit Sarkaria; Dec 2023) — Q-learning with lookup tables and a neural network written from scratch.
- **Image classification with ensemble stacking** (EECE 571T, Matt Yedlin; Apr 2023) — best three-model CNN stack 88.7% vs best single model 85.6% on CIFAR-10.
- **Feature extractors for few-shot learning** (EECE 570, Xiaoxiao Li; Apr 2023) — ResNet-18, DenseNet-121 and ConvNeXt-T as Prototypical Network backbones on CUB-200 (83.8 / 81.8 / 78.6%).
- **Speech re-synthesis from MRI with a graph convolutional network** (EECE 571F, Renjie Liao; Dec 2022) — MFCC encoding; GCN on electromagnetic articulography data; MAE and mel-cepstral distortion.
- **Unsupervised brain-tumour segmentation** (EECE 562, Zhen Wang; Dec 2022) — GMM-EM from scratch in **PyTorch** on BraTS 2018; Dice ≈ 0.9 on six of ten patients without labels.
- **Denoising diffusion-weighted MRI with PCA and DCT** (ELEC 523, Siamak Najarian; Dec 2022) — principally responsible for implementation (**NumPy**); ~5 dB SNR gain (PCA) vs ~1.7 dB (DCT).

TECHNICAL SKILLS

- **Languages:** Python, Go, TypeScript / JavaScript, Java, SQL
- **Backend & data:** FastAPI, SQLAlchemy, Alembic, Pydantic, Go Fiber, GORM, Spring Boot, REST API design, JWT / RBAC, PostgreSQL / PostGIS, MySQL, MongoDB Atlas, Redis, GeoPandas / Shapely, NetworkX, schema & ERD design, data contracts, data validation frameworks
- **LLM & AI:** PyTorch, NumPy, SciPy, HuggingFace, OpenAI API, RAG, vector search, prompt engineering & prefix caching, rule-first LLM pipelines, LLM evaluation, diffusion models, knowledge distillation, few-shot learning, Slurm
- **Frontend, cloud & testing:** React 19, TypeScript, TanStack Query, Redux, Tailwind CSS, Pytest, JUnit / Mockito, Postman, AWS (Elastic Beanstalk, EC2, CloudWatch), Docker Compose, Git / GitHub PR workflow